

Cross-Patient Neural Speech Decoding

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1 What Is Shared Across Patients?

Conserved cortical organization

Speech motor cortex has a **conserved somatotopic layout** across all humans: face, jaw, tongue, and larynx representations are arranged in a consistent order along the central sulcus (Breshears 2015, $n=33$; Bouchard 2013; Roux 2020). The *ordinal* arrangement is invariant; the *absolute positions* vary by 5–15 mm between individuals.

We define 16 **cortical nodes** at atlas-derived positions (Brainnetome, $N=40$ HCP subjects) corresponding to known speech-relevant regions. These are the subject-invariant functional units:

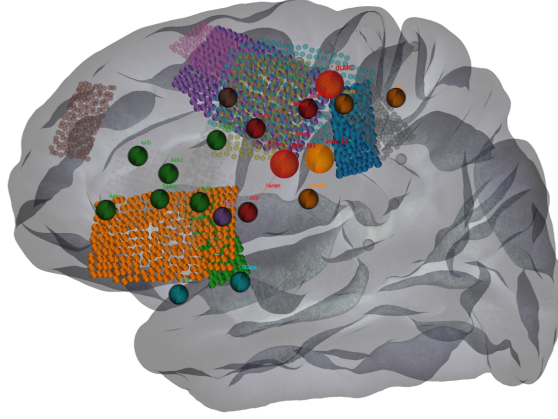


Figure 1: 16 cortical nodes (large spheres) and patient electrode arrays (small dots). Each patient covers a different subset of nodes.

Key nodes along the motor strip (dorsal $\rightarrow$ ventral): **A4hf** (face M1, MNI $-49/-6/40$) $\rightarrow$ **A6cvl** (ventral PMC, $-49/6/31$) $\rightarrow$ **A4tl** (tongue M1, $-52/2/9$). Plus 3 sensory (face/tongue S1, proprioceptive), 6 Broca's (articulatory planning), 2 auditory (self-monitoring), 1 insula, 1 prefrontal. Each node reachable by 3–9 of our 10 patients.

Subject-invariant phoneme dynamics

Each phoneme activates a **specific, predictable combination of cortical nodes** determined by articulatory biomechanics:

- **/b/, /p/** (bilabial stops): lip closure $\rightarrow$ **face M1** fires
- **/g/, /k/** (velar stops): tongue dorsum raises $\rightarrow$ **tongue M1** fires
- **/v/** (labiodental fricative): lower lip to teeth $\rightarrow$ **face M1** fires
- **/a/** (open vowel): jaw opening + tongue low $\rightarrow$ **tongue M1** + jaw area
- **/i/** (high front vowel): tongue raised front $\rightarrow$ **tongue M1**

- /u/ (high back vowel): tongue raised back + lip rounding → **tongue M1 + face M1**

These node activations follow a conserved temporal cascade: **Broca's** (−200 to −50 ms, planning) → **Motor cortex** (−50 to +100 ms, execution) → **Somatosensory** (+50 to +200 ms, feedback) → **Auditory** (+100 to +300 ms, self-monitoring).

The ground truth: what is conserved across patients is (1) the existence and gross positions of cortical nodes representing articulatory muscles, and (2) the spatiotemporal activation sequences across these nodes during phoneme production. A 3-phoneme token like /abe/ produces a characteristic trajectory through the node network — Broca's plans the sequence, then face M1 fires for /b/, tongue M1 for /a/ and /e/, with sensory and auditory feedback following each.

2 What Do We Observe?

We have **noisy, spatially-sampled observations** of these node activations. Each patient has a μ ECoG electrode array (128 or 256 channels, 1.3–1.7 mm pitch) pressed against the cortical surface during intra-operative speech. We measure high-gamma activity (HGA, 70–150 Hz envelope at 200 Hz) — a proxy for local cortical population activity.

The physics of this measurement is well-characterized: the signal at each electrode is a **linear superposition** of cortical sources, contributions **decay with distance** ($\sim 1/d^2$), and sources beyond ~ 25 mm are below the noise floor. Crucially, this spatial mapping depends **only on geometry** — it does not change based on what the brain is currently doing.

Every patient's array sits in a **different position** (15–25 mm offsets, variable rotation). Each array is a partial view of the cortical node network from a different vantage point.

3 What Must We Learn?

Per patient (geometry — *where* things are)

1. **Electrode localization error** (6 params): Surgical reconstruction has ~ 4 –8 mm systematic rigid-body error. Learn translation Δ + rotation ω , clamped ($\|\Delta\| \leq 15$ mm, $\|\omega\| \leq 0.15$ rad).
2. **Individual cortical node positions** (48 params): Individual anatomy varies by 5–15 mm from atlas. Each of 16 nodes gets a learned offset $\delta_l \in \mathbb{R}^3$, clamped ≤ 15 mm. Initialized at zero (trust the atlas).
3. **Signal gain and offset** (128 params): Electrode impedance varies. Per-channel scale (positive via softplus) and offset. Initialized from channel statistics.

Total: 182 parameters per patient. Comparable cross-patient iEEG systems (see significant, MIBRAIN, BarISTA) use per-subject heads or region prototypes with similar or larger per-patient footprints, or no per-patient adaptation at all.

Shared (dynamics — *what* things mean)

Spatiotemporal activation patterns on cortical nodes → which phoneme sequence. Given correctly-positioned nodes with correctly-calibrated signals pooled into them, the 16-node network dynamics encode the phoneme. Same mapping for every patient — determined by conserved cortical organization, not individual anatomy.

4 What Architecture Does This Ask For?

Each component follows directly from the physics:

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Raw electrode HGA (N_i electrodes x T time steps)
-> Temporal binning: Conv1d(1->d, k=10, s=10)           [200Hz -> 20Hz]
-> Per-patient calibration:                               [182 params/pt]
    softplus(s) * x + b   (gain/offset, positive)
    correct electrode positions (6 params: rigid)
    correct node positions (48 params: individual somatotopy)
-> Spatial pooling into 16 cortical nodes                 [physics-derived]
    Wendland kernel, distance-weighted, compact support
-> [Node self-attention + Temporal self-attention] x 2   [learns dynamics]
-> AR decoder: predict phoneme 1, 2, 3 + beam search     [52 valid tokens]

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Spatial pooling

The electrode-to-node mapping is a **source localization** problem. The physics constrains it to be: **linear** (superposition), **distance-determined** (not content-dependent), **compactly supported** (sources beyond ~ 25 mm are below the noise floor), and **instantaneous**.

Each node aggregates nearby electrode signals using distance-weighted spatial pooling. Each attention head has a learned receptive field radius R_h and a learned feature projection ($x \mapsto xW_h$), selecting *which* temporal features to aggregate at *which* spatial scale:

$$z_{\text{node}}^{(h)} = \sum_j \frac{w(d_{lj}, R_h)}{\sum_{j'} w(d_{lj'}, R_h)} \cdot (x_j W_h) \quad (1)$$

where $w(d, R)$ is a smooth, compactly-supported kernel (exactly zero beyond R). Spatial weights are purely geometric; feature transformation is purely functional.

Parameter budget

Component	Params	Note
Conv1d + feature projections	$\sim 5\text{K}$	Shared: temporal features
Spatial pooling (Wendland)	H scalars	Shared: learned R_h per head
Spatiotemporal blocks ($\times 2$)	$\sim 132\text{K}$	Shared: node dynamics
AR decoder	$\sim 4\text{K}$	Shared: phoneme sequence
Gain/offset ($\text{softplus}(s) \odot x + b$)	128/pt	Per-patient: impedance
Electrode correction (Δ, ω)	6/pt	Per-patient: registration error
Node offsets (δ_l)	48/pt	Per-patient: individual somatotopy
Total shared	$\sim 141\text{K}$	
Total per-patient	182	

Immediate next steps

1. **Scale data** (highest priority) — to learn the shared spatiotemporal dynamics we need a foundation of data. Could request chronic ECoG speech data from Flinker/Chen (NYU, 48 patients) and/or Chang (UCSF, ~ 15 – 25 patients)
2. **Nonlinear MNI normalization** — current electrode transform is linear-only (~ 5 – 15 mm residual error); if T1 MRIs exist, ANTs SyN could halve this
3. **Implementation** — architecture is designed; target ~ 2 weeks to first training run on DCC